

Genetic Algorithms for Estimating Attribute Importance and Interaction in Soybean Crop Yield Prediction

Daniel Natan dos Santos Brito

Advisor: Prof. Renato Tinós

Department of Computing and Mathematics, Faculty of Philosophy, Sciences and Letters of Ribeirão Preto, University of São Paulo

danielnatanbrito@usp.br

Objectives

The project aims to apply the genetic algorithm with linkage learning (GAwLL) [1] to estimate the importance and interaction of features used by different machine learning models in predicting soybean crop yield, a topic already addressed using climate data and vegetation indices [2]. In addition to comparing the predictive performance of multilayer perceptrons (MLP), decision trees, random forests, and K-nearest neighbors (KNN), it seeks to identify, through GAwLL, which agronomic variables contribute most to the prediction and how they relate to one another—information that is relevant both for model explainability and for understanding the agricultural problem itself.

Methods and Procedures

GAwLL is based on a standard genetic algorithm (GA), in which each individual $\mathbf{x}$ is a binary vector of size N that indicates which of the N available features compose the solution, evaluated by a fitness function $f(\mathbf{x})$. For each evaluation, GAwLL generates three individuals derived from $\mathbf{x}$: one with the g -th bit flipped, another with the h -th bit flipped, and a third with both flipped simultaneously. From the fitness differences among these variations, the following are calculated:

$$\delta_g(\mathbf{x}) = f(\mathbf{x} \oplus \mathbf{1}_g) - f(\mathbf{x}) \quad (1)$$

$$\delta_g(\mathbf{x} \oplus \mathbf{1}_h) = f(\mathbf{x} \oplus \mathbf{1}_h \oplus \mathbf{1}_g) - f(\mathbf{x} \oplus \mathbf{1}_h) \quad (2)$$

$$\omega_{g,h}(\mathbf{x}) = |\delta_g(\mathbf{x} \oplus \mathbf{1}_h) - \delta_g(\mathbf{x})| \quad (3)$$

where $\oplus$ represents the XOR operation and $\mathbf{1}_g$ is a binary vector with bit 1 only at the g -th position. Two

variables x_g and x_h interact when $\omega_{g,h}(\mathbf{x}) > 0$. The importance of a variable g , in turn, is given by the maximum absolute value of δ_g observed among the solutions $\mathbf{x}$ visited in the search space $Y(g)$:

$$\hat{v}_{imp}(g) = \max |\delta_g(\mathbf{x})|, \mathbf{x} \in Y(g) \quad (4)$$

In addition, the interaction between the variables x_g and x_h is given by the maximum value of $\omega_{g,h}(\mathbf{x})$:

$$\hat{v}_{int}(g, h) = \max \omega_{g,h}(\mathbf{x}), \mathbf{x} \in Y(g, h) \quad (5)$$

The central point of GAwLL is that this importance and interaction information is obtained as a byproduct of the GA optimization itself, without requiring additional fitness evaluations beyond those already performed during the search. To this end, a Python implementation of GAwLL was used, extending its original application, which was limited to KNN, to multiple machine learning models [3].

For the experiments, a dataset from the ABC Foundation was used, comprising 3,675 instances and 54 physical, chemical, biological, and management features related to soybean crops, with yield (kg/hectare) as the target. The four models (MLP, Decision Tree, Random Forest, and KNN) were trained with default Scikit-learn hyperparameters and evaluated through 5-fold cross-validation using accuracy as the metric. The genetic algorithm was configured with a population of 100 individuals, 100 generations, a crossover rate of 0.8, a mutation rate of 0.3, roulette wheel selection, and elitism, with the following fitness function:

$$f(\mathbf{x}) = 0,9 \cdot \text{acc}(\mathbf{x}) + 0,1 \cdot \left(1 - \sum_{i=1}^n x_i / N\right) \quad (6)$$

which combines the accuracy (acc) of the model with the attributes given by $\mathbf{x}$ and with a penalty term proportional to the number of selected features, encouraging solutions with high accuracy and few features.

Results

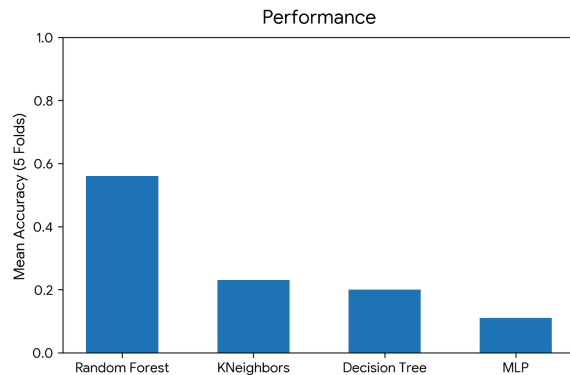


Figure 1: Mean accuracy (5 folds) for each model with all 54 features.

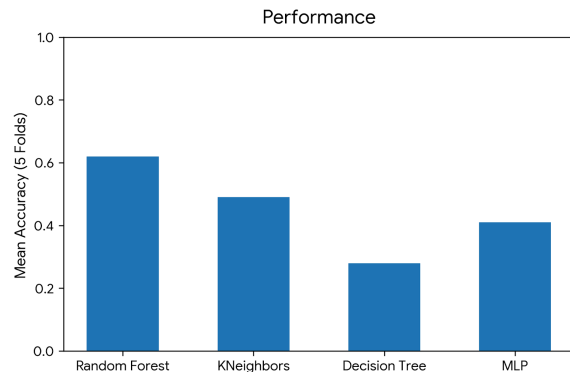


Figure 2: Mean accuracy (5 folds) for each model with the attributes selected by GAWLL .

Figures 1 and 2 show the accuracy results from training with the original 54 features, where Random Forest achieved the highest mean accuracy (≈ 0.56), followed by KNN (≈ 0.23), Decision Tree (≈ 0.20), and MLP (≈ 0.11). All models improved after GAWLL selection, with notable gains for Random Forest ($0.56 \rightarrow 0.62$) and MLP ($0.11 \rightarrow 0.41$). The average number of selected features was 13, representing a reduction of approximately 76% compared to the original feature set.

Among the features identified as most important by GAWLL, key variables include field location (latitude and longitude), genotype and its purpose, and phytosanitary management variables such as herbicide, fungicide, and insecticide applications, which appeared among the selected features across most models.

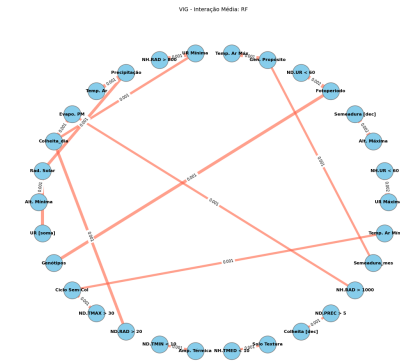


Figure 3: Mean interaction between features in the Random Forest model

Figure 3 shows the interaction between features in the Random Forest model, highlighting interactions such as sowing 10-day period and maximum altitude (0.002), as well as the number of days with mean relative humidity < 60% and photoperiod (0.002).

Conclusions

The results show that a large portion of the original features contributes little to yield prediction, and that models trained using only the features selected by GAWLL not only maintained but surpassed the performance achieved with the full set of variables. This suggests that using all features may mislead the models, likely due to the presence of noise or uninformative variables.

Acknowledgements

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References

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